

Zirui (Ken) Deng

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SUMMARY

PhD candidate in Computer Science with expertise in machine learning, statistical modeling, and data-driven systems.

Experienced in building deep learning models and scalable data pipelines for time-series and structured data.

Motivated in applying ML to large-scale recommendation, marketplace, and optimization problems.

Technical Tools: Python, MATLAB, R, Java, C/C++, AWS, SQL, Git, Linux, Bash, Excel, LaTeX

Soft Skills: Cross-functional Collaboration, Project Management, Multi-tasking, Innovation, Presentation

EDUCATION

Washington University in St. Louis

St. Louis, Missouri

PhD Candidate in Computer Science

August 2021 – May 2027 (anticipated)

• Research interest: theoretical and applied methods in information theory, quantitative modeling, and machine learning

Emory University

Atlanta, Georgia

B.S. in Applied Mathematics and Statistics & B.S. in Computer Science (Major GPA: 3.92/4.0)

August 2017 – May 2021

• Coursework: Numerical Analysis, Calculus, Linear Algebra, Optimization, Partial Differential Equations, Probability & Statistics, Data Structures and Algorithms, Database Systems, Systems Programming, Deep Learning, Machine Learning

RESEARCH EXPERIENCE

Privacy Certification for ML Models via PMC

St. Louis, Missouri

Washington University in St. Louis

August 2025 – Present

- Designing a unified privacy certification framework leveraging Pointwise Maximum Cost (PMC), enabling quantification of membership inference risk in modern machine learning models.
- Developing scalable estimation pipelines linking empirical attack metrics to theoretical guarantees, supporting production-grade ML model monitoring and reducing evaluation latency.
- Analyzing adversarial strategies across models (SVMs, generative models), deriving optimal inference policies and improving robustness of ML systems under distribution shift.

Efficient Vector Symbolic Architectures via Histogram Recovery

St. Louis, Missouri

Washington University in St. Louis

February 2025 – August 2025

- Developed coding-theoretic vector symbolic architectures (VSAs) for neurosymbolic AI using concatenated Reed-Solomon and Hadamard codes, enabling efficient compositional representation with formal and provable quasi-orthogonality and recovery guarantees.
- Designed efficient superposition-recovery algorithms via a novel histogram recovery formulation, leveraging list-decoding and lattice decoding techniques to recover noisy compositional representations without heuristic search or training.
- Constructed noise-resilient linear-code-based VSAs supporting efficient encoding, binding recovery, and scalable retrieval of compositional objects, improving recovery guarantees relative to prior random-code and Hadamard-based approaches. (Preprint link: <https://arxiv.org/abs/2511.01838>)

Private Inference in Quantized Models

St. Louis, Missouri

Washington University in St. Louis

September 2022 – May 2025

- Developed information-theoretic private inference protocols for quantized ML models, enabling secure retrieval of inference signals while protecting both user data and proprietary model parameters. Designed PIR (Private Information Retrieval)-inspired query-answer frameworks for private inner-product computation.
- Derived optimal privacy-communication tradeoffs and lower bounds on information leakage under polynomial decoding constraints. Proposed protocols for inference over binary and quantized weights using coding theory and combinatorics.
- Extended private inference schemes to jointly retrieve multiple neural-network activation signals for scalable privacy-preserving deep learning inference. (Publication link: <https://ieeexplore.ieee.org/abstract/document/10896693>)

Perfect Subset Privacy in Polynomial Computation

St. Louis, Missouri

Washington University in St. Louis

October 2023 – April 2025

- Developed information-theoretic privacy-preserving protocols for distributed polynomial computation, enabling secure delegation to untrusted service providers with provable subset privacy guarantees. Designed Reed-Muller coded computing schemes with reduced communication overhead and robustness to worker stragglers.

- Generalized subset-private encoding schemes to arbitrary finite fields using linear-code constructions and mutual information guarantees. Developed communication-efficient recovery protocols based on Reed-Muller information sets and polynomial interpolation.
 - Investigated the concept and construction of information super-sets for Reed-Muller codes to construct straggler-resilient distributed computation frameworks.
- (Publication link: <https://ieeexplore.ieee.org/document/10619107>)

Predicting Surgical Case Duration Using Neural ODEs

Washington University in St. Louis

St. Louis, Missouri

August 2021 – November 2021

- Designed & implemented a time-series prediction pipeline, integrating LSTM and Neural ODE models to capture both discrete and continuous dynamics in heterogeneous clinical data.
- Developed robust data preprocessing workflows for multimodal datasets (static + temporal features), improving training stability and enabling scalable experimentation across RNN/LSTM-based architectures.
- Optimized numerical solvers and training procedures for stability and efficiency, reducing surgical duration prediction error by 15% (CRPS) and improving reliability in real-world healthcare forecasting tasks.

Undergraduate Honors thesis: Fair algorithms in machine learning

Emory University

Atlanta, Georgia

May 2020 – May 2021

- Analyzed algorithmic fairness in ML classification and clustering models, defining fairness via algorithmic bias in ML-based decisions on sensitive variables (e.g., gender, ethnicity, sexual orientation).
- Developed and evaluated optimization-based fairness constraints (e.g., parity regularization, constrained ERM), reducing statistical parity gap with <5% loss in predictive accuracy.
- Evaluated fairness-utility tradeoffs under distributional shifts, showing improved out-of-sample generalization with reduced disparity and stable predictive performance on held-out test sets.

Multi-variate Neural Ordinary Differential Equations for Time-series Modeling

Remote from Emory University

Atlanta, Georgia

May 2020 – July 2020

- Designed and implemented deep learning models (CNN, LSTM, Neural ODE) for time-series medical data, capturing discrete and continuous dynamics and reducing forecasting error by ~12% (MSE) relative to baseline RNN models.
- Developed data augmentation and contrastive learning strategies to improve representation quality, achieving 8-10% improvement in downstream prediction accuracy and matching state-of-the-art sequence modeling benchmarks.
- Optimized training pipelines for stability and scalability (batching, learning rate scheduling, numerical stability), reducing training time and enabling efficient experimentation across large-scale time-series datasets.

INTERNSHIPS

Data Analyst Internship

China Telecom

Nanjing, China

May 2019 – June 2019

- Developed automated data pipelines using Python, Selenium, Pandas, enabling large-scale web scraping (10K+ records/day) and improving data collection efficiency for financial analysis workflows.
- Implemented validation and cleaning scripts (Python, NumPy) to ensure data integrity, reducing data inconsistencies and lowering manual processing time.
- Contributed to a financial robotics project by building modular data processing components, supporting near real-time ingestion and improving downstream analytics latency by ~30%.

Investment Banking Analysis Internship

Ten Billion Research Laboratory in Shanghai

Shanghai, China

May 2018 – June 2018

- Conducted quantitative market research on blockchain technologies using Python, Excel, analyzing 50+ projects/datasets and identifying key industry trends through SWOT and time-series analysis.
- Developed forecasting insights from structured and unstructured data, improving prediction accuracy of short-term market trends by ~20% compared to baseline heuristic models.
- Presented data-driven findings in cross-functional meetings, delivering actionable investment insights that informed strategy discussions and improved decision turnaround time by ~25%.

ADDITIONAL INFORMATION

Languages: Fluent in English and Mandarin Chinese

Interests: Poker; Weiqi (amateur 5-dan, national second-level athlete)